



- 7 (a) Fig. 7.1 shows two resistors connected in series with a cell of electromotive force (e.m.f.) 1.50 V and internal resistance $0.28\ \Omega$.

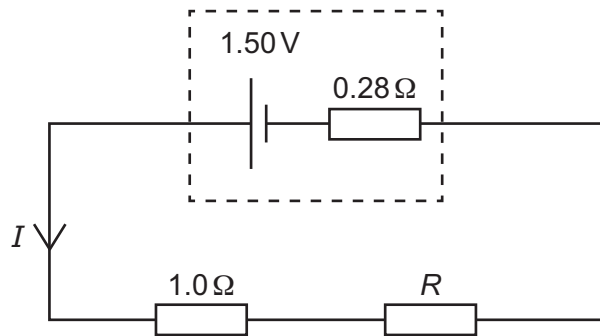


Fig. 7.1

One of the resistors has resistance $1.0\ \Omega$. The other resistor has resistance R . The terminal potential difference (p.d.) across the cell is 1.36 V.

- (i) Show that the current I in the circuit is 0.50 A.

[2]

- (ii) Calculate the combined resistance of the two resistors.

resistance = Ω [2]

- (iii) Use your answer in (a)(ii) to determine resistance R .

$R =$ Ω [1]



(b) The circuit in Fig. 7.1 is disconnected and the two resistors are reconnected to the cell, now in parallel with each other.

(i) On Fig. 7.2, complete the circuit diagram to show this arrangement.

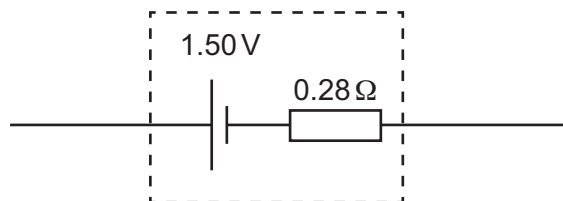


Fig. 7.2

[1]

(ii) Explain, without calculation, whether the terminal p.d. across the cell is now less than, equal to or greater than 1.36 V.

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..... [2]

[Total: 8]





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